ICT2113 · Data Structures and Algorithms

Practical Tutorial: Graph Traversal

Original synthetic reading for demonstration and independent study. It is not an official syllabus, lecture note, or assessment specification.

Learning objectives

After this reading, a learner should be able to explain the central idea, apply it to a small example, and identify at least one assumption or limitation.

1. Main idea

A graph models vertices and edges. Breadth-first search explores a queue of frontier vertices and finds shortest paths by edge count in an unweighted graph. Depth-first search explores a branch before backtracking and is useful for reachability and cycle analysis.

2. Example and reasoning

Represent a course prerequisite network as adjacency lists. Start from one course, mark it visited before enqueuing, and record each vertex predecessor. Marking before enqueueing prevents duplicate frontier entries when several edges reach the same vertex.

3. A method for practice

1. State the problem and define the terms that affect the answer.
2. Write down the assumptions and information available.
3. Work through a small example and make each step visible.
4. Check the result against a second case or independent explanation.
5. Summarize what the result supports and what it does not establish.

4. Questions for study

- Explain “practical tutorial: graph traversal” in your own words and give a new example.
- Which assumption matters most in the example, and how would the answer change if it were false?
- Describe one common mistake and a practical way to detect it.
- Write a short checklist another learner could use to review this work.

Reflection

Separate evidence from assumptions. If a conclusion depends on information not provided here, identify what would need to be verified.